Consolidated Resolution Packet for Problem 3.19 and Theorem 3.21

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Source Problem and Consolidation Status

The source is Jan Rozendaal and Mark Veraar, *Fourier multiplier theorems involving type and cotype*, arXiv:1605.09340. Problem 3.19 asks:

> Let $1 \le p \le 2 \le q \le \infty$ and $r \in (1, \infty]$ be such that
> $$\frac{1}{r} = \frac{1}{p} - \frac{1}{q}.$$
> Classify those Banach spaces X and Y for which
> $$T_m \in \mathcal{L}(L^p(\mathbb{R}^d; X), L^q(\mathbb{R}^d; Y))$$
> for all X-strongly measurable maps $m : \mathbb{R}^d \to \mathcal{L}(X, Y)$ such that
> $$\{|\xi|^{d/r} m(\xi) : \xi \in \mathbb{R}^d \setminus \{0\}\}$$
> is γ-bounded.

In Theorem 3.21 we provide conditions under which one can take $p = p_0$ and $q = q_0$. The general case we state as an open problem:

Problem 3.19. *Let $1 \le p \le 2 \le q \le \infty$ and $r \in (1, \infty]$ be such that $\frac{1}{r} = \frac{1}{p} - \frac{1}{q}$. Classify those Banach spaces X and Y for which $T_m \in \mathcal{L}(L^p(\mathbb{R}^d; X), L^q(\mathbb{R}^d; Y))$ for all X-strongly measurable maps $m : \mathbb{R}^d \to \mathcal{L}(X, Y)$ such that $\{|\xi|^{d/r} m(\xi) : \xi \in \mathbb{R}^d \setminus \{0\}\}$ is γ-bounded.*

The same problem can be formulated in case m is scalar-valued, in which case

This packet consolidates the previous run artifacts for arXiv:1605.09340 into one authoritative packet. It contains:

- the complete scalar classification of Problem 3.19;
- endpoint obstructions for every nonzero Banach pair X, Y;
- the repaired form of Theorem 3.21 and the proof-gap explanation for its printed $p = 1$ endpoint;
- the latest interior obstruction and literature map.

The status is *partial result, likely valid.* The packet does not claim a full classification of the remaining interior Banach-valued problem.

Proof Intuition

There are two separate phenomena.

First, all endpoint failures are scalar. A nonzero pair X, Y contains a rank-one scalar channel: choose $x^* \in X^*$, $x_0 \in X$, and $y_0 \in Y$, and set

$$m(\xi)x = a(\xi)x^*(x)y_0.$$

If $|\xi|^{d/r}a(\xi)$ is bounded, then the corresponding operator family is γ-bounded. Testing on $f(t) = h(t)x_0$ reduces the Banach-valued estimate to the scalar multiplier a. The scalar endpoint failure is the classical strong endpoint failure of the Riesz potential: approximate identities see the kernel singularity $|x|^{-d/q}$ and produce logarithmic L^q blowup. The $q = \infty$ face follows by duality from the same even Riesz multiplier.

Second, the corrected positive theorem uses a Sobolev–γ bridge. The source proof factors through

$$\dot{H}_p^{d/p-d/2}(\mathbb{R}^d; X) \hookrightarrow \gamma(\mathbb{R}^d; X), \qquad \gamma(\mathbb{R}^d; Y) \hookrightarrow \dot{H}_q^{d/q-d/2}(\mathbb{R}^d; Y).$$

These embeddings are available in the source paper's non-endpoint type/cotype range and in the sharp p-convex/q-concave lattice setting for $1 < p \le 2 \le q < \infty$. At $p = 1$ the first embedding would imply a false strong endpoint Sobolev/Riesz estimate. That is the small but decisive gap in the printed endpoint of Theorem 3.21.

The Scalar Classification

Theorem 1 (Scalar form of Problem 3.19)**.** *Let $d \ge 1$. In the scalar case $X = Y = \mathbb{C}$, the property in Problem 3.19 holds for the admissible exponents $1 \le p \le 2 \le q \le \infty$, $r \in (1, \infty]$, $1/r = 1/p - 1/q$, if and only if*

$$1 < p \le 2 \le q < \infty.$$

Proof. If $1 < p \le 2 \le q < \infty$, scalar γ-boundedness is just uniform boundedness. Hence

$$|m(\xi)| \le C|\xi|^{-d/r}.$$

For $r < \infty$, this implies $m \in L^{r,\infty}(\mathbb{R}^d)$, and the scalar Hörmander multiplier theorem quoted in the source paper gives $T_m : L^p(\mathbb{R}^d) \to L^q(\mathbb{R}^d)$. If $r = \infty$, then $p = q = 2$, and the conclusion follows from Plancherel.

Now let $p = 1$, $2 \le q < \infty$, and $r = q'$. Put

$$a_q(\xi) = |\xi|^{-d/q'} \qquad (\xi \ne 0),$$

with any value assigned at $\xi = 0$. Then $|\xi|^{d/q'}a_q(\xi) = 1$ for $\xi \ne 0$. The inverse Fourier transform of a_q is a nonzero constant multiple of

$$K_q(x) = |x|^{-d/q}.$$

Let $\eta \in C_c^\infty(\mathbb{R}^d)$ be nonnegative, supported in the unit ball, and satisfy $\int \eta = 1$. Set $\eta_\varepsilon(x) = \varepsilon^{-d}\eta(x/\varepsilon)$. For $2\varepsilon \le |x| \le 1/2$ and $y \in \operatorname{supp}(\eta_\varepsilon)$, one has $|x - y| \simeq |x|$. Therefore

$$|T_{a_q}\eta_\varepsilon(x)| \ge c|x|^{-d/q}$$

on this annulus, and hence

$$\|T_{a_q}\eta_\varepsilon\|_q^q \ge c^q \int_{2\varepsilon \le |x| \le 1/2} |x|^{-d}\, dx \simeq \log(1/\varepsilon) \to \infty.$$

But $\|\eta_\varepsilon\|_1 = 1$. Thus T_{a_q} is not bounded $L^1 \to L^q$.

Finally let $1 < p \le 2$ and $q = \infty$. Then $r = p$. If the scalar property held for this pair, it would hold for the real even multiplier

$$a_p(\xi) = |\xi|^{-d/p}.$$

For Schwartz f, g, Fourier-side pairing gives

$$\langle T_{a_p} f, g\rangle = \langle f, T_{a_p} g\rangle.$$

Boundedness $T_{a_p} : L^p \to L^\infty$ would imply

$$|\langle f, T_{a_p} g\rangle| \le C\|f\|_p \|g\|_1.$$

Taking the supremum over $\|f\|_p \le 1$ yields $T_{a_p} : L^1 \to L^{p'}$, contradicting the $p = 1$ endpoint failure just proved with target exponent p'. This proves the scalar classification. $\square$

Endpoint Obstructions for Nonzero Banach Pairs

Theorem 2 (Endpoint obstruction for all nonzero pairs)**.** *Let $d \ge 1$, and let X, Y be nonzero Banach spaces.*

1. *If $p = 1$, $2 \le q < \infty$, and $r = q'$, then Problem 3.19 fails for the pair X, Y.*
2. *If $1 < p \le 2$, $q = \infty$, and $r = p$, then Problem 3.19 fails for the pair X, Y.*

Proof. Choose $x_0 \in X$, $x^* \in X^*$, $y_0 \in Y$, and $y^* \in Y^*$ such that

$$x^*(x_0) = 1, \qquad y_0 \ne 0, \qquad y^*(y_0) = 1.$$

For $p = 1$, $2 \le q < \infty$, let $a_q(\xi) = |\xi|^{-d/q'}$ and define

$$m(\xi)x = a_q(\xi)x^*(x)y_0.$$

The weighted family is γ-bounded. Indeed, for finite choices $\xi_j \ne 0$, $u_j \in X$, and $\lambda_j = |\xi_j|^{d/q'} a_q(\xi_j)$, one has $|\lambda_j| = 1$, and

$$\begin{aligned}\Big(\mathbb{E}\Big\|\sum_j \gamma_j \lambda_j x^*(u_j) y_0\Big\|_Y^2\Big)^{1/2} &\le \|y_0\| \Big(\sum_j |x^*(u_j)|^2\Big)^{1/2} \\ &\le \|y_0\|\,\|x^*\| \Big(\mathbb{E}\Big\|\sum_j \gamma_j u_j\Big\|_X^2\Big)^{1/2}.\end{aligned}$$

Testing on $f_\varepsilon = \eta_\varepsilon x_0$ gives

$$T_m f_\varepsilon = (T_{a_q}\eta_\varepsilon) y_0.$$

By Theorem 1, the $L^q(Y)$-norms diverge while $\|f_\varepsilon\|_{L^1(X)}$ remains bounded. Thus the $p = 1$ endpoint fails.

For $1 < p \le 2$, $q = \infty$, define

$$m_p(\xi)x = |\xi|^{-d/p} x^*(x) y_0.$$

The same rank-one estimate shows that $\{|\xi|^{d/p} m_p(\xi) : \xi \ne 0\}$ is γ-bounded. If $T_{m_p} : L^p(\mathbb{R}^d; X) \to L^\infty(\mathbb{R}^d; Y)$ were bounded, then testing on $f x_0$ and applying y^* would imply the scalar boundedness

$$T_{a_p} : L^p(\mathbb{R}^d) \to L^\infty(\mathbb{R}^d), \qquad a_p(\xi) = |\xi|^{-d/p},$$

contradicting the scalar $q = \infty$ failure in Theorem 1. Hence this endpoint fails for every nonzero pair X, Y. $\square$

Corollary 1 (Remaining exponent core)**.** *For nonzero X, Y, the only exponent range in Problem 3.19 where a nontrivial positive classification can remain is*

$$1 < p \leq 2 \leq q < \infty.$$

The formal corner $p = 1, q = \infty$ is not admissible in Problem 3.19, since it would force $r = 1$, while the problem assumes $r \in (1, \infty]$.

Theorem 3.21: Printed Endpoint Conflict and Repair

The source paper's Theorem 3.21 is printed for

$$p \in [1, 2], \qquad q \in [2, \infty),$$

and says that, under p-convex/q-concave lattice hypotheses, the same weighted γ-boundedness assumption implies $T_m : L^p(\mathbb{R}^d; X) \to L^q(\mathbb{R}^d; Y)$.

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each $j \in \mathcal{J}$, $\{|\xi|^{\frac{d}{r}} m_j(\xi) \mid \xi \in \mathbb{R}^d\} \subseteq \mathcal{L}(X, Y)$ is γ-bounded by C. Note that, since X and Y have finite cotype, γ-boundedness and R-boundedness are equivalent. Now we claim that $\{\widetilde{T_{m_j}} \mid j \in \mathcal{J}\} \subseteq \mathcal{L}(L^p(\mathbb{R}^d; X), L^q(\mathbb{R}^d; Y))$ is γ-bounded as well. To prove this claim one can use the method of [22, Theorem 3.2]. Indeed, using their notation, it follows from the Kahane-Khintchine inequalities that $\mathrm{Rad}(X)$ has the same type as X and $\mathrm{Rad}(Y)$ has the same cotype as Y. Therefore, given $j_1, \ldots, j_n \in \mathcal{J}$ and the corresponding $m_{j_1}, \ldots, m_{j_n}$, one can apply Theorem 3.18 to the multiplier $M : \mathbb{R}^d \setminus \{0\} \to \mathcal{L}(\mathrm{Rad}(X), \mathrm{Rad}(Y))$ given as the diagonal operator with diagonal $(m_{j_1}, \ldots, m_{j_n})$. In order to check the γ-boundedness one now applies property (α) as in [22, Estimate (3.2)].

3.5. **Convexity, concavity and L^p-L^q results in lattices.** In this section we will prove certain sharp results in p-convex and q-concave Banach lattices.

First of all, from the proof of Theorem 3.18 we obtain the following result with the sharp exponents p and q.

Theorem 3.21. *Let $p \in [1, 2]$, $q \in [2, \infty)$, and let $r \in [1, \infty]$ be such that $\frac{1}{r} = \frac{1}{p} - \frac{1}{q}$. Let X be a complemented subspace of a p-convex Banach lattice with finite cotype and Y a Banach space that is continuously embedded in a q-concave Banach lattice. Let $m : \mathbb{R}^d \to \mathcal{L}(X, Y)$ be an X-strongly measurable map such that $\{|\xi|^{\frac{d}{r}} m(\xi) \mid \xi \in \mathbb{R}^d \setminus \{0\}\} \subseteq \mathcal{L}(X, Y)$ is γ-bounded. Then T_m extends uniquely to a bounded map $\widetilde{T_m} \in \mathcal{L}(L^p(\mathbb{R}^d; X), L^q(\mathbb{R}^d; Y))$ with*

$$\|\widetilde{T_m}\|_{\mathcal{L}(L^p(\mathbb{R}^d;X), L^q(\mathbb{R}^d;Y))} \leq C\gamma(\{|\xi|^{\frac{d}{r}} m(\xi) \mid \xi \in \mathbb{R}^d \setminus \{0\}\}), \tag{3.11}$$

where C is a constant depending on X, Y, p, q and d.

Proof. In the case where X is a p-convex and Y is a q-concave Banach lattice, the

The literal $p = 1$ endpoint conflicts with Theorem 2: take $X = Y = \mathbb{C}$. More concretely, at $p = 1, q = 2, r = 2$, the scalar multiplier

$$m(\xi) = \mathbf{1}_{\{|\xi| \geq 1\}} |\xi|^{-d/2}$$

has bounded weighted set $\{|\xi|^{d/2}m(\xi) : \xi \neq 0\}$, but approximate identities and Plancherel give logarithmic L^2-blowup. The high-frequency cutoff merely removes any possible concern about the multiplier at $\xi = 0$; the endpoint obstruction is the same Riesz singularity.

The proof passage for Theorem 3.21 says that the embeddings used in Theorem 3.18 can be proved for p-convex and q-concave lattices and then imports the proof of Theorem 3.18.

Proof. In the case where X is a p-convex and Y is a q-concave Banach lattice, the embeddings in (3.10) can be proved in the same way as in [60, Theorem 3.9], where the inhomogeneous case was considered. Therefore, the result in this case follows from the proof of Theorem 3.18.

Now let X_0 be a p-convex Banach lattice with finite cotype such that $X \subseteq X_0$, let $P \in \mathcal{L}(X_0)$ be a projection with range X and let Y_0 be a q-concave Banach lattice with a continuous embedding $\iota : Y \hookrightarrow Y_0$. Let $m_0 : \mathbb{R}^d \to \mathcal{L}(X_0, Y_0)$ be given by $m_0(\xi) := \iota \circ m(\xi) \circ P \in \mathcal{L}(X_0, Y_0)$ for $\xi \in \mathbb{R}^d$. It is easily checked that $\{m_0(\xi) \mid \xi \in \mathbb{R}^d\} \subseteq \mathcal{L}(X_0, Y_0)$ is γ-bounded, with

(3.12)
$$\gamma(\{m_0(\xi) \mid \xi \in \mathbb{R}^d \setminus \{0\}\}) \leq \|\iota\|_{\mathcal{L}(Y,Y_0)}\|P\|_{\mathcal{L}(X_0)}\gamma(\{|\xi|^{\frac{d}{r}}m(\xi) \mid \xi \in \mathbb{R}^d \setminus \{0\}\}).$$

As we have shown above, there exists a constant $C \in (0, \infty)$ that depends only on X_0, Y_0, p, q and d such that T_{m_0} extends uniquely to a bounded operator $\widetilde{T_{m_0}} \in \mathcal{L}(L^p(\mathbb{R}^d; X_0), L(\mathbb{R}^d; Y_0))$ with

$$\|T_{m_0}\|_{\mathcal{L}(L^p(\mathbb{R}^d;X_0),L^q(\mathbb{R}^d;Y_0))} \leq C\gamma(\{m_0(\xi) \mid \xi \in \mathbb{R}^d\}). \tag{3.13}$$

Since $T_m = \widetilde{T_{m_0}}\restriction_{\mathcal{S}(\mathbb{R};X)}$, the result follows from (3.12) and (3.13). □

Remark 3.22. Note from (3.12) and (3.13) that the constant C in (3.11) depends on X and Y as $C = \|P\|_{\mathcal{L}(X_0)}\,\|\iota\|_{\mathcal{L}(Y,Y_0)}\,C_1$, where $P \in \mathcal{L}(X_0)$ is a projection with range X on a p-convex Banach lattice X_0 with finite cotype, $\iota \in \mathcal{L}(Y, Y_0)$ is a

At $p = 1$, that embedding would imply a false strong endpoint Sobolev/Riesz estimate. For $1 < p \leq 2$, the same argument is valid.

Theorem 3 (Repaired Theorem 3.21). *Let $1 < p \leq 2$, $2 \leq q < \infty$, and let $r \in (1, \infty]$ satisfy*

$$\frac{1}{r} = \frac{1}{p} - \frac{1}{q}.$$

Let X be a complemented subspace of a p-convex Banach lattice with finite cotype, and let Y be a Banach space that embeds continuously into a q-concave Banach lattice. If $m : \mathbb{R}^d \setminus \{0\} \to \mathcal{L}(X, Y)$ is X-strongly measurable and

$$\{|\xi|^{d/r}m(\xi) : \xi \neq 0\}$$

is γ-bounded, then T_m extends uniquely to a bounded operator

$$T_m : L^p(\mathbb{R}^d; X) \to L^q(\mathbb{R}^d; Y),$$

with norm bounded by a constant depending on X, Y, p, q, d times the displayed γ-bound.

Proof. In the lattice case the homogeneous embedding input is

$$\dot{H}_p^{d/p-d/2}(\mathbb{R}^d; X) \hookrightarrow \gamma(\mathbb{R}^d; X), \qquad \gamma(\mathbb{R}^d; Y) \hookrightarrow \dot{H}_q^{d/q-d/2}(\mathbb{R}^d; Y).$$

For $1 < p \leq 2 \leq q < \infty$, this is exactly the non-endpoint p-convex/q-concave embedding input cited by the source proof, based on Veraar's embedding results for γ-spaces and the Kalton–van Neerven–Veraar–Weis Besov-to-γ embeddings.

Set

$$m_1(\xi) = |\xi|^{d/2-d/p}, \qquad m_2(\xi) = |\xi|^{d/r} m(\xi) m_1(\xi).$$

For $f \in \mathcal{S}(\mathbb{R}^d; X)$, the source proof gives

$$\begin{aligned}
\|T_m f\|_{L^q(\mathbb{R}^d;Y)} &= \|T_{m_2} f\|_{\dot{H}_q^{d/q-d/2}(\mathbb{R}^d;Y)} \\
&\le C\|T_{m_2} f\|_{\gamma(\mathbb{R}^d;Y)} \\
&\le C\,\gamma(\{|\xi|^{d/r} m(\xi) : \xi \neq 0\})\|m_1 \widehat{f}\|_{\gamma(\mathbb{R}^d;X)} \\
&\le C\,\gamma(\{|\xi|^{d/r} m(\xi) : \xi \neq 0\})\|T_{m_1} f\|_{\dot{H}_p^{d/p-d/2}(\mathbb{R}^d;X)} \\
&= C\,\gamma(\{|\xi|^{d/r} m(\xi) : \xi \neq 0\})\,\|f\|_{L^p(\mathbb{R}^d;X)}.
\end{aligned}$$

The middle estimate is the γ-multiplier theorem. Density of Schwartz functions in L^p gives the bounded extension.

For complemented subspaces and embedded targets, let $X \subseteq X_0$ be the range of a bounded projection $P : X_0 \to X$, where X_0 is a p-convex Banach lattice with finite cotype, and let $\iota : Y \to Y_0$ be a continuous embedding into a q-concave Banach lattice. Apply the lattice case to

$$m_0(\xi) = \iota\, m(\xi) P \in \mathcal{L}(X_0, Y_0).$$

The weighted family for m_0 has γ-bound at most

$$\|\iota\|\,\|P\|\,\gamma(\{|\xi|^{d/r} m(\xi) : \xi \neq 0\}).$$

Restricting back to X proves the stated theorem. □

Corollary 2. *The $p = 1$ endpoint in the printed Theorem 3.21 cannot be rescued for nonzero spaces under only the stated weighted γ-boundedness hypothesis. The sharp corrected strong range is $1 < p \le 2$, $2 \le q < \infty$.*

Interior Range: What Remains Open

After Corollary 1, the possible nontrivial range for Problem 3.19 is

$$1 < p \le 2 \le q < \infty.$$

The source paper proves a large sufficient theorem: if X has type $p_0 \in (1, 2]$, Y has cotype $q_0 \in [2, \infty)$, and

$$1 < p < p_0, \qquad q_0 < q < \infty,$$

with the Hilbert endpoint allowances when $p_0 = 2$ or $q_0 = 2$, then Problem 3.19 has a positive answer. The repaired Theorem 3 gives a sharp class in the lattice setting.

The best abstract reduction is the following: a full positive theorem in the expected type/cotype form would follow from the endpoint Bessel–γ embeddings

$$\dot{H}_p^{d/p-d/2}(\mathbb{R}^d; X) \hookrightarrow \gamma(\mathbb{R}^d; X), \qquad \gamma(\mathbb{R}^d; Y) \hookrightarrow \dot{H}_q^{d/q-d/2}(\mathbb{R}^d; Y).$$

The decisive known theorem of Kalton–van Neerven–Veraar–Weis characterizes type and cotype by the adjacent Besov embeddings

$$B_{p,p}^{d(1/p-1/2)}(\mathbb{R}^d; X) \hookrightarrow \gamma(L^2(\mathbb{R}^d), X)$$

and

$$\gamma(L^2(\mathbb{R}^d), Y) \hookrightarrow B^{d(1/q-1/2)}_{q,q}(\mathbb{R}^d; Y).$$

For $p < 2$, the Besov space $B^s_{p,p}$ is smaller than the Bessel/Triebel–Lizorkin space $H^s_p = F^s_{p,2}$ needed in the multiplier proof. Dually, the cotype conclusion lands in the wrong summability scale for the target Bessel embedding. This is the main remaining summability gap.

The Rozendaal–Veraar Besov companion paper proves optimal type/cotype exponents in the Besov scale and derives an $L^p \to L^q$ theorem under the dyadic condition

$$\left(2^{kd/r}\gamma(\{m(\xi) : \xi \in J_k\})\right)_{k\in\mathbb{Z}} \in \ell^1(\mathbb{Z}).$$

This condition is stronger than Problem 3.19's uniform weighted gamma-bound, which only gives a bounded dyadic sequence.

Dominguez–Veraar later developed vector-valued Pitt inequalities. Their endpoint theorem gives the right Bochner estimate in key cases, for example

$$\left\| |\xi|^{-d(1/p-1/2)} \widehat{f}(\xi) \right\|_{L^2(\mathbb{R}^d;X)} \lesssim \|f\|_{L^p(\mathbb{R}^d;X)}$$

when X has suitable Fourier type. This is not the same as the gamma-radonifying embedding above; passing from Bochner $L^2(X)$ to $\gamma(L^2, X)$ requires additional square-function geometry, such as the Hilbert or lattice structures already covered.

Finally, the source paper's Schur multiplier problem asks whether a Schatten-class endpoint $1/r = |1/a - 1/2|$ can be reached. The authors note that a negative answer would imply failure of

$$H^{1/a-1/2}_a(\mathbb{R}; S^a) \to \gamma(\mathbb{R}; S^a), \qquad 1 < a < 2.$$

This is the same Bessel–γ endpoint obstruction. Modern Littlewood–Paley–Stein and Lusin type/cotype results are adjacent to this issue, but the literature search did not locate a theorem converting those square-function estimates into the global radonifying embedding required here.

Verification Notes

- The scalar positive range uses only the scalar weak-L^r Hörmander theorem quoted in the source paper. The case $r = \infty$ is $p = q = 2$ and follows from Plancherel.
- The scalar $p = 1$ failure is the standard endpoint Riesz-potential failure; the logarithmic divergence is localized on $2\varepsilon \le |x| \le 1/2$.
- The $q = \infty$ failure uses Schwartz-pairing duality and does not identify the dual of L^∞.
- The rank-one reduction proves that the endpoint failures are not artifacts of $X = Y = \mathbb{C}$; they occur for every nonzero pair.
- The Theorem 3.21 repair deletes exactly the endpoint contradicted by the rank-one obstruction. For $1 < p \le 2$, the proof is the source paper's Sobolev–γ multiplier argument.
- The interior material is recorded as an obstruction/reduction, not as a proof of the full classification.

Literature and Duplicate Checks

The run indexes were checked for `1605.09340`, Problem 3.19, Theorem 3.21, scalar classification, endpoint repair, proof gap, Pitt inequality, type/cotype, Besov multiplier, and Sobolev–γ phrases. The separate earlier artifacts have been consolidated into this packet.

External literature checks covered the source paper, the Besov companion arXiv:1606.03272, Kalton–van Neerven–Veraar–Weis arXiv:math/0610620, Dominguez–Veraar arXiv:1904.07930, and modern Littlewood–Paley–Stein/Lusin type papers including arXiv:2105.12175 and arXiv:2401.13932. No full later classification of Problem 3.19 or erratum for Theorem 3.21 was found in this search.

Superseded Run Artifacts

This packet supersedes the following previous run artifacts:

- `solutions/partial/1605.09340_scalar_endpoint_multiplier_failure/`
- `solutions/partial/1605.09340_theorem_3_21_endpoint_repair/`
- `solutions/partial/1605.09340_problem_3_19_endpoint_core_reduction/`
- `proof_gaps/1605.09340_theorem_3_21_scalar_endpoint_conflict/`
- `attempts/1605.09340_problem_3_19_interior_literature_full_push.md`

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Human Review Recommendation

Review the Riesz-potential endpoint proof, the rank-one γ-boundedness estimate, and the use of the $p > 1$ Sobolev–γ embedding in the repaired theorem. If these pass, this single packet should replace the earlier separate scalar, proof-gap, repair, endpoint-core, and interior-attempt records. The remaining interior classification should stay marked open/partial unless a proof or counterexample for the endpoint Bessel–γ mechanism is found.

Review note. *This is a consolidation packet. It intentionally preserves the full mathematical content of the earlier packets while leaving only one active durable record for arXiv:1605.09340 in the run indexes.*